

Day 09 – Technology Fingerprinting Report

Trios Cyber Internship

Assignment: Technology Fingerprinting

Day: 09

Target: Damn Vulnerable Web Application (DVWA)

Target IP: 192.168.56.101

Target URL: http://192.168.56.101/dvwa/

Environment: Authorized Local VirtualBox Laboratory

Tools: WhatWeb, curl, Burp Suite

1. Objective

The objective of this exercise was to perform technology fingerprinting against an authorized local web application.

The assessment focused on identifying technologies and components exposed through HTTP responses, response headers, and application content.

The exercise included:

- Running WhatWeb against DVWA.
 - Reviewing HTTP response headers using curl.
 - Identifying the web server and version.
 - Identifying operating system information.
 - Identifying PHP and its version.
 - Identifying the DVWA application.
 - Reviewing cookies and redirect behavior.
 - Building a technology inventory.
 - Collecting screenshots as supporting evidence.
-

2. Scope

Testing was performed only against the intentionally vulnerable DVWA instance hosted inside the authorized local VirtualBox laboratory.

Target IP: 192.168.56.101

Target URL: http://192.168.56.101/dvwa/

No external systems were targeted.

3. Tools Used

WhatWeb

WhatWeb was used to identify technologies and application characteristics exposed by the target.

Command:

```
“‘bash whatweb -a 3 http://192.168.56.101/dvwa/”
```

4. Initial HTTP Response Analysis

The target was tested using curl to manually inspect the HTTP response headers.

Command used:

```
curl -noproxy '*' -I -max-time 5 http://192.168.56.101/dvwa/
```

The target returned an HTTP 302 Found response.

Important information observed:

HTTP/1.1 302 Found Server: Apache/2.2.8 (Ubuntu) DAV/2 X-Powered-By: PHP/5.2.4-2ubuntu5.10
Set-Cookie: PHPSESSID=... Set-Cookie: security=high Location: login.php Content-Type: text/html

The 302 status indicates that the requested resource redirected the client to another location. The Location header identified login.php as the redirect destination.

5. WhatWeb Fingerprinting Results

The following command was executed:

```
whatweb -a 3 http://192.168.56.101/dvwa/
```

WhatWeb identified the following technologies and characteristics:

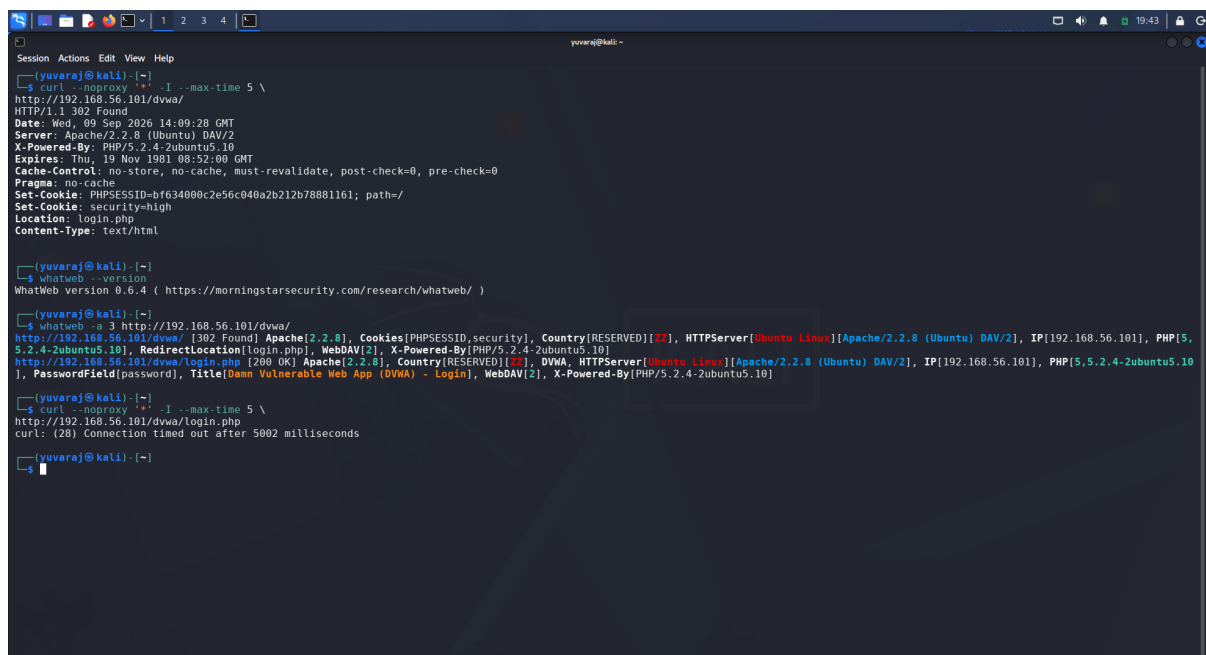
- Apache 2.2.8
- Ubuntu Linux
- PHP 5.2.4
- PHP package version 5.2.4-2ubuntu5.10
- WebDAV 2
- Damn Vulnerable Web App (DVWA)
- PHPSESSID cookie
- Security cookie
- X-Powered-By header
- Redirect to login.php

WhatWeb also identified the DVWA login page:

<http://192.168.56.101/dvwa/login.php>

The login page was identified as Damn Vulnerable Web App and returned HTTP 200 OK.

Evidence



```
(yuvraj@kali) ~$ curl -noproxy '*' -I -max-time 5 \
http://192.168.56.101/dvwa/
HTTP/1.1 302 Found
Date: Wed, 09 Sep 2026 14:09:28 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Cache-Control: no-store, no-cache, must-revalidate, post-check=0, pre-check=0
Pragma: no-cache
Set-Cookie: PHPSESSID=b7634000c2e56c040a2b212b78881161; path=/
Set-Cookie: security=high
Location: login.php
Content-Type: text/html

(yuvraj@kali) ~$ whatweb -v
WhatWeb version 0.6.4 ( https://morningstarsecurity.com/research/whatweb/ )

(yuvraj@kali) ~$ whatweb -a 3 http://192.168.56.101/dvwa/
http://192.168.56.101/dvwa/ [302 Found] Apache[2.2.8], Cookies[PHPSESSID,security], Country[RESERVED][??], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[192.168.56.101], PHP[5.2.4-2ubuntu5.10], RedirectLocation[login.php], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]
http://192.168.56.101/dvwa/login.php [200 OK] Apache[2.2.8], Country[RESERVED][??], DVWA, HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[192.168.56.101], PHP[5.2.4-2ubuntu5.10], PasswordField[password], Title[Damn Vulnerable Web App (DVWA) - Login], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]

(yuvraj@kali) ~$ curl -noproxy '*' -I -max-time 5 \
http://192.168.56.101/dvwa/login.php
curl: (28) Connection timed out after 5002 milliseconds

(yuvraj@kali) ~$
```

Figure 1: WhatWeb Technology Fingerprint

6. Manual HTTP Response Header Review

The HTTP response headers were manually reviewed using curl.

Command used:

```
curl -noproxy '*' -I -max-time 5 http://192.168.56.101/dvwa/
```

Server Header

Server: Apache/2.2.8 (Ubuntu) DAV/2

This disclosed the Apache web server, its version, Ubuntu Linux information, and WebDAV support.

X-Powered-By Header

X-Powered-By: PHP/5.2.4-2ubuntu5.10

This disclosed PHP as the server-side scripting technology and exposed the PHP version.

Set-Cookie Headers

The server returned PHPSESSID and security cookies.

PHPSESSID indicates PHP-based session management.

The security=high cookie reflected the DVWA security level observed during the assessment.

Location Header

Location: login.php

This indicated that the initial request was redirected to the DVWA login page.

Content-Type Header

Content-Type: text/html

This indicated that the server returned HTML content.

Evidence

7. DVWA Login Page Inspection

The DVWA login page was inspected directly using curl.

Command used:

```
curl -noproxy '*' -s -max-time 5 http://192.168.56.101/dvwa/login.php | head -30
```

The returned HTML contained the following title:

Damn Vulnerable Web App (DVWA) - Login

A focused search was also performed using:

```
curl -noproxy '*' -s http://192.168.56.101/dvwa/login.php | grep -iE '
|powered|generator|version'
```

The output confirmed:

Damn Vulnerable Web App (DVWA) - Login

This confirmed the identity of the web application as Damn Vulnerable Web App.

```
Session Actions Edit View Help
(yuvaraj@kali) ~
$ curl -s -o /dev/null -w '%{httpcode} %{http_header}' -I --max-time 5 \
http://192.168.56.101/dvwa/
HTTP/1.1 302 Found
Date: Wed, 09 Sep 2026 14:09:28 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Cache-Control: no-store, no-cache, must-revalidate, post-check=0, pre-check=0
Pragma: no-cache
Set-Cookie: PHPSESSID=bf634000c2e56c040a2b212b78881161; path=/
Set-Cookie: security=high
Location: login.php
Content-Type: text/html

(yuvaraj@kali) ~
$ whatweb --version
WhatWeb version 0.6.4 ( https://morningstarsecurity.com/research/whatweb/ )

(yuvaraj@kali) ~
$ whatweb -a 3 http://192.168.56.101/dvwa/
http://192.168.56.101/dvwa/ [302 Found] Apache[2.2.8], Cookies[PHPSESSID,security], Country[RESERVED][22], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[192.168.56.101], PHP[5.2.4-2ubuntu5.10], RedirectLocation[login.php], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]
http://192.168.56.101/dvwa/login.php [200 OK] Apache[2.2.8], Country[RESERVED][22], DVWA, HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[192.168.56.101], PHP[5.2.4-2ubuntu5.10], PasswordField[password], Title[Damn Vulnerable Web App (DVWA) - Login], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]

(yuvaraj@kali) ~
$ curl -s -o /dev/null -w '%{httpcode} %{http_header}' -I --max-time 5 \
http://192.168.56.101/dvwa/login.php
curl: (28) Connection timed out after 5002 milliseconds

(yuvaraj@kali) ~
$ curl -s -o /dev/null -w '%{httpcode} %{http_header}' -I --max-time 5 \
http://192.168.56.101/dvwa/login.php
curl: (7) Failed to connect to 192.168.56.101:80 after 3079 ms: Could not connect to server

(yuvaraj@kali) ~
$
```

Figure 2: HTTP Response Headers

Evidence

8. Technology Inventory

Category	Identified Technology / Information	Evidence
Web Application	Damn Vulnerable Web App (DVWA)	WhatWeb / HTML title
Web Server	Apache 2.2.8	HTTP response / WhatWeb
Operating System	Ubuntu Linux	WhatWeb / Server header
Server Module	WebDAV 2	Server header / WhatWeb
Server-Side Technology	PHP 5.2.4	X-Powered-By / WhatWeb
PHP Package Version	5.2.4-2ubuntu5.10	X-Powered-By
Session Management	PHPSESSID	Set-Cookie
DVWA Security Level	High	Security cookie
Authentication Endpoint	login.php	Location header
Response Format	text/html	Content-Type
Application Title	DVWA Login	HTML title

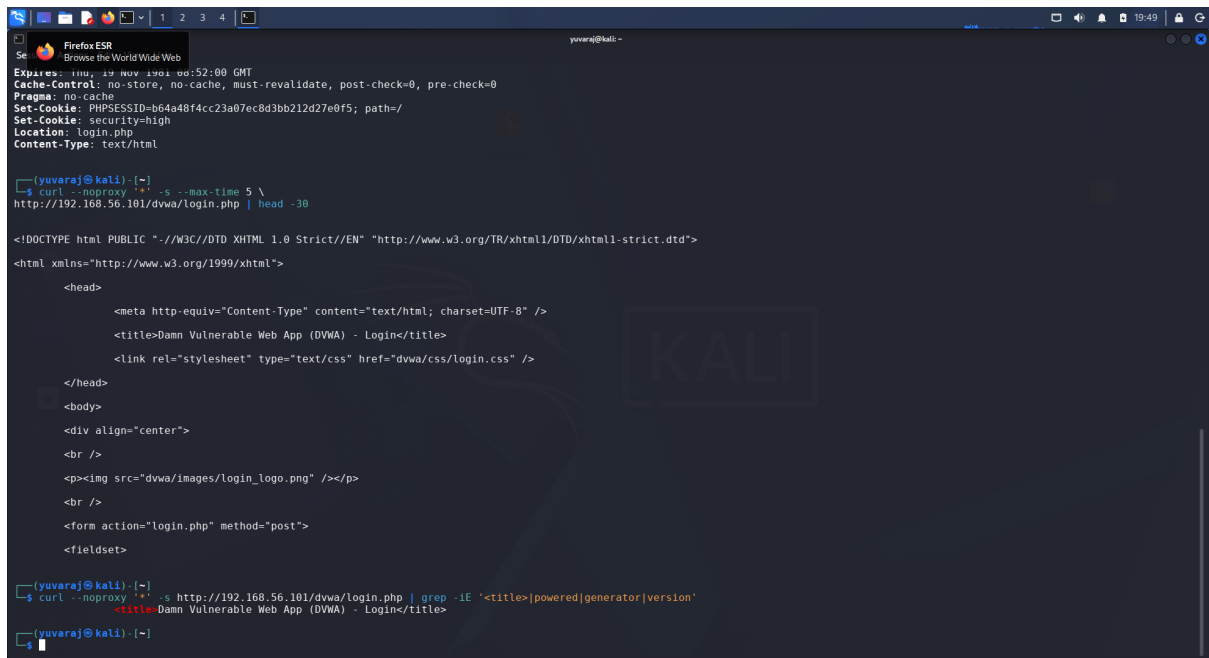
9. Key Findings

Finding 1 – Apache Version Disclosure

The target disclosed the Apache web server version 2.2.8.

Finding 2 – Ubuntu Linux Disclosure

WhatWeb identified Ubuntu Linux as the operating system associated with the web server.



```
(yuvraj@kali) ~  
$ curl --no-proxy '*' -s --max-time 5 \ http://192.168.56.101/dvwa/login.php | head -30  
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">  
<html xmlns="http://www.w3.org/1999/xhtml">  
  <head>  
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />  
    <title>Damn Vulnerable Web App (DVWA) - Login</title>  
    <link rel="stylesheet" type="text/css" href="dvwa/css/login.css" />  
  </head>  
  <body>  
    <div align="center">  
      <br />  
      <p></p>  
      <br />  
      <form action="login.php" method="post">  
        <fieldset>
```

Figure 3: Technology Details

Finding 3 – PHP Version Disclosure

The X-Powered-By header disclosed PHP version 5.2.4-2ubuntu5.10.

Finding 4 – WebDAV Support Disclosure

The Server header contained DAV/2, indicating WebDAV support.

Finding 5 – Application Identification

WhatWeb identified the application as Damn Vulnerable Web App (DVWA).

The login page title also confirmed the application identity.

Finding 6 – Session Management

The server issued a PHPSESSID cookie, indicating PHP session management.

Finding 7 – Authentication Redirect

The /dvwa/ endpoint redirected unauthenticated requests to login.php.

10. Security Observations

Technology fingerprinting is useful during security assessments because applications and servers may disclose implementation details through HTTP headers and application responses.

The target disclosed:

- Web server software
- Web server version
- Operating system information
- PHP runtime
- PHP version
- WebDAV support
- Application identity

- Session-management information
- Authentication endpoint

The Apache and PHP versions identified during the assessment are legacy software versions and should be reviewed in a modern production environment.

The Server and X-Powered-By headers also expose technology and version information.

From a defensive perspective, unnecessary technology and version disclosure should generally be minimized on production systems.

11. Learning Outcomes

This exercise provided practical experience with:

1. Using WhatWeb for web technology fingerprinting.
2. Reviewing HTTP response headers using curl.
3. Identifying web server software and versions.
4. Identifying operating system information.
5. Identifying server-side scripting technologies.
6. Identifying web application technologies.
7. Understanding HTTP status codes and redirects.
8. Recognizing information disclosure through HTTP headers.
9. Building a structured technology inventory.
10. Collecting screenshots as supporting evidence.
11. Documenting technical security findings.
12. Understanding the importance of minimizing unnecessary technology disclosure.

12. Evidence Summary

Evidence	File	Purpose
Screenshot 1	01-whatweb-fingerprint.png	WhatWeb technology fingerprinting
Screenshot 2	02-http-response-headers.png	Manual HTTP response header analysis
Screenshot 3	03-technology-details.png	Application and technology details

All three screenshots were captured during the authorized laboratory exercise and copied into the Day 09 project directory.

13. Conclusion

The Day 09 Technology Fingerprinting assessment was successfully completed against the authorized DVWA laboratory target.

WhatWeb identified Apache 2.2.8, Ubuntu Linux, PHP 5.2.4, WebDAV 2, and the Damn Vulnerable Web App application.

Manual HTTP header analysis using curl confirmed the server information, PHP version, session cookies, DVWA security cookie, redirect behavior, and content type.

The assessment demonstrated how technology fingerprinting can help security professionals understand a web application's technology stack and identify information that may require additional security review.

The exercise also highlighted the security importance of reviewing information disclosure through HTTP headers and application responses.

14. Scope and Ethics

All testing was performed against the intentionally vulnerable DVWA application hosted inside the authorized local VirtualBox laboratory environment.

Target:

192.168.56.101

No external systems were targeted.

The assessment was limited to technology identification, HTTP response header review, application-content inspection, and documentation as required by the Day 09 assignment.

15. Final Status

Day 09 – Technology Fingerprinting: COMPLETED

The required technology fingerprinting, manual HTTP header review, technology inventory, evidence collection, and technical documentation were completed successfully.